Experience with B200 and B300 GPUs on the DSAI Arch Cluster

3/16/2026

Code and results are under /scratch/dzarzhi1/ext-ndtrung and /scratch/dzarzhi1/ext-zand

Test Run Workloads (Trung)

1. **Pangu ML training** (3D Vision Transformer) (PI Pedram Hassanzadeh)
   - training code from Dr. Youzhi Yu's repo (RCC), subset of the training data from PI group
   - *Metrics: samples/second (higher is better)*
2. **Simulated ML benchmark**
   - Operations: GEMM, Conv2D, DDP, All-reduce, P2P, dev-dev, host-device transfers (link)
   - *Metrics: TFLOPs and GB/s (higher is better)*
3. **GROMACS 2025.3**
   - Standard input: STMV (1 million atoms (https://developer.nvidia.com/blog/massively-improved-multi-node-nvidia-gpu-scalability-with-gromacs/)
   - single precision (FP32)
   - *Metrics: nanoseconds/days (higher is better)*
4. **LAMMPS 11Feb2026**
   - Standard input: LJ (2M – 32M particles)
   - GPU package in mixed precision (FP64 for force and energy accumulation, FP32 for atom position loads and intermediate arithmetic ops)
   - *Metrics: timesteps/second (higher is better)*

Representative AI Workloads (Mohsen)

1. Computer Vision (CV) Training

- ResNet50 (CNN Baseline): Represents traditional deep learning workloads. Used to measure the efficiency of convolutional operations and peak floating-point throughput.
- Vision Transformer (ViT_L_16): Represents the modern "Attention-based" architecture. This is a memory-intensive workload used to test the high-bandwidth memory (HBM3e) capabilities.
- *Metrics: Throughput (FPS): Measuring total raw productivity [Higher is Better].*

2. Large Language Model (LLM) Inference

- Llama-3-8B (Dense Model): A high-throughput, latency-sensitive model used to measure "Tokens Per Second" (TPS) and the efficiency of the Tensor Cores in BF16, FP8, and FP4 precisions.
- Llama-3-70B (Large-Scale Model): A high-capacity model used to test Consolidation (running a 70B model on a single B300 card) vs. Parallelization (sharding across 4 or 8 GPUs using Tensor Parallelism).
- Engine: Powered by the vLLM V1 Engine, simulating production-grade inference environments on the DSAI cluster.
- *Metrics: Throughput (TPS): Measuring total raw productivity [Higher is Better] and Responsiveness (TTFT/Latency): Measuring Time to First Token [Lower is Better].*

Pangu ML training of 3D Vision Transformer

		H100* (samples/sec)	B200 (samples/sec)	B300 (samples/sec)
Precision (with 4 GPUs & batch size = 8)	BF16	3.72	4.04	2.21
	FP16	3.66	4.22	2.18
	FP32 /w TF32	2.38	3.72	2.14
	FP32	1.64	1.88	1.41
	FP8	N/A		2.22
Batch size (with 4 GPUs and FP16)	4	3.28	5.05	
	8	3.57	4.84	2.66
	12	3.41	5.08	2.26
Number of GPUs (FP16 & batch size = 2 per GPU)	1	1.22	1.67	1.13
	2	2.20	2.95	1.79
	4	3.63	4.90	2.26

/scratch/dzarzhi1/ext-ndtrung/pangus2s/dsai/b300_training_dsai.sh
* obtained on the RCC Midway

The higher the result, the better
B300 GPU performance is surprisingly lower than for B200 GPU

Simulated ML benchmark

https://github.com/rcc-uchicago/nodes-testing-tools/tree/main/ml-training/gpu_benchmark.py

	B200	B300
Attention (Transformer, BERT-like)		
Attention B32 H12 S512 D64 (torch.float16)	87.32 TFLOPS	62.53 TFLOPS
Conv2D (ResNet-style)		
Conv2D 128x128x112x112 k3 (torch.float16)	349.28 TFLOPS	345.95 TFLOPS
Data Parallel Training:		
Data Parallel 1024x1024x1024 (torch.float16) on 4 GPUs	15.99 TFLOPS	1.24 TFLOPS
Data Parallel 4096x4096x4096 (torch.float16) on 4 GPUs	1033.57 TFLOPS	82.09 TFLOPS
Collective Communication (All-Reduce):		
All-Reduce 100 MB on 4 GPUs	472.06 GB/s	251.73 GB/s
All-Reduce 500 MB on 4 GPUs	510.73 GB/s	491.80 GB/s
Peer-to-Peer Transfer:		
P2P Transfer 100 MB (bidirectional)	452.20 GB/s	357.76 GB/s
P2P Transfer 500 MB (bidirectional)	600.05 GB/s	617.57 GB/s

Synthetic benchmark 4x B300 GPUs

GEMM Performance

Test	Actual	Min Expected	Max Expected	Status
GEMM bfloat16 small_4096x4096x4096	782.95	1750.00	2150.00	⚠️ BELOW
GEMM bfloat16 large_8192x8192x8192	1632.22	1950.00	2350.00	⚠️ BELOW
GEMM bfloat16 xlarge_16384x16384x16384	1576.07	2050.00	2400.00	⚠️ BELOW

Device-to-Device Performance

Test	Actual	Min Expected	Max Expected	Status
Device-to-Device Copy 512 MB	1547.59	7000.00	8000.00	⚠️ BELOW
Device-to-Device Copy 1024 MB	3031.14	7000.00	8000.00	⚠️ BELOW
Device-to-Device Copy 2048 MB	2893.19	7000.00	8000.00	⚠️ BELOW

All-Reduce & Peer-to-peer (P2P) Performance

Test	Actual	Min Expected	Max Expected	Status
All-Reduce 10 MB on 4 GPUs	131.61	155.00	195.00	⚠️ BELOW
All-Reduce 100 MB on 4 GPUs	523.71	155.00	195.00	✓ ABOVE
All-Reduce 500 MB on 4 GPUs	457.19	155.00	195.00	✓ ABOVE
P2P Transfer 10 MB (GPU 0 <-> GPU 1)	88.87	1250.00	1650.00	⚠️ BELOW
P2P Transfer 100 MB (GPU 0 <-> GPU 1)	454.38	1250.00	1650.00	⚠️ BELOW
P2P Transfer 500 MB (GPU 0 <-> GPU 1)	479.43	1250.00	1650.00	⚠️ BELOW

/scratch/dzarzhi1/ext-ndtrung/ML-training/gpu_benchmark.py

Synthetic benchmark 4x B200 GPUs

GEMM Performance

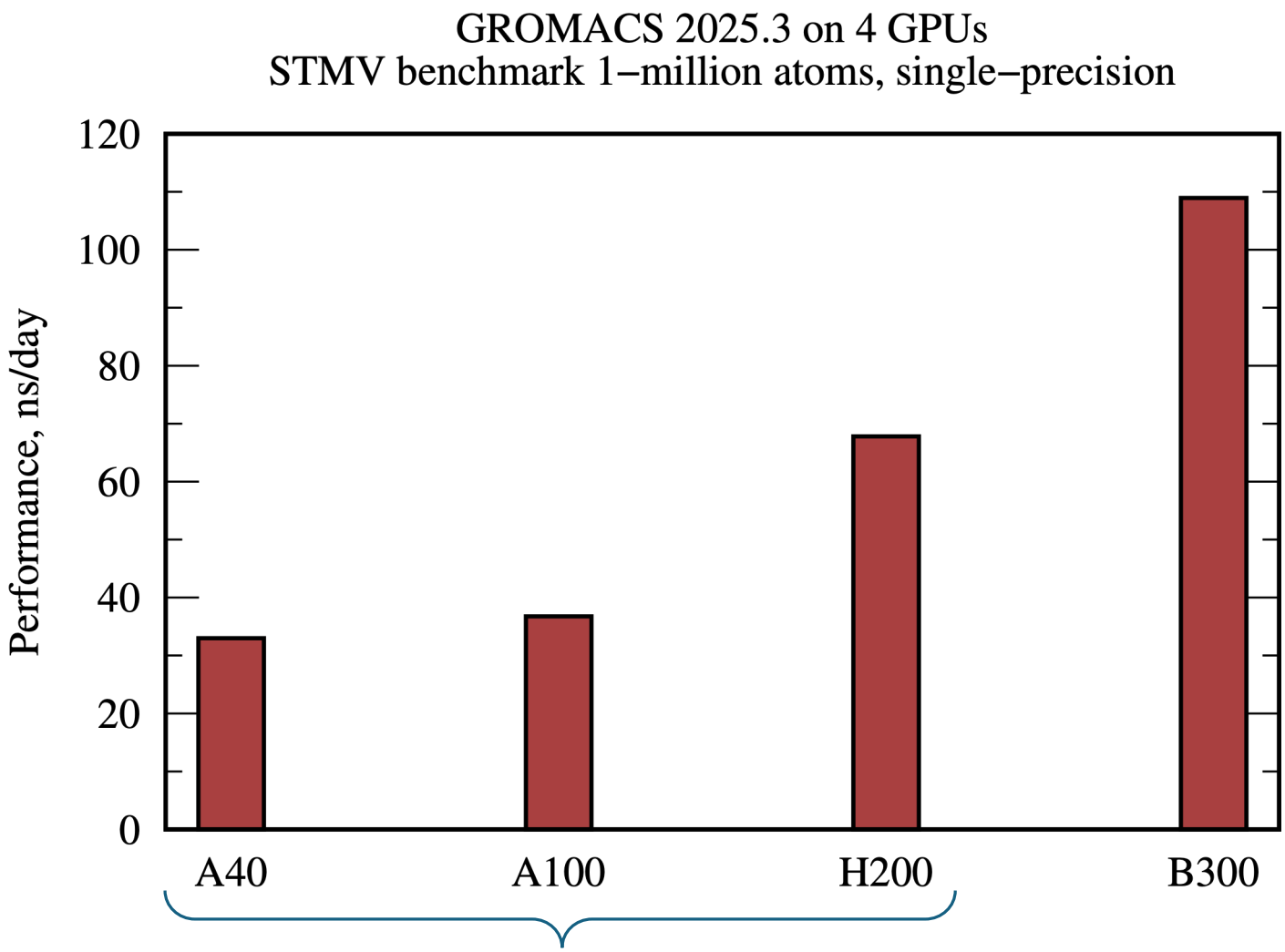
Test	Actual	Min Expected	Max Expected	Status
GEMM bfloat16 small_4096x4096x4096	1504.25	1600.00	2000.00	⚠️ BELOW
GEMM bfloat16 large_8192x8192x8192	1405.09	1800.00	2200.00	⚠️ BELOW
GEMM bfloat16 xlarge_16384x16384x16384	1511.55	1900.00	2250.00	⚠️ BELOW

Device-to-Device Performance

Test	Actual	Min Expected	Max Expected	Status
Device-to-Device Copy 512 MB	2937.60	7000.00	8000.00	⚠️ BELOW
Device-to-Device Copy 1024 MB	3002.98	7000.00	8000.00	⚠️ BELOW
Device-to-Device Copy 2048 MB	3033.43	7000.00	8000.00	⚠️ BELOW

All-Reduce & Peer-to-peer (P2P) Performance

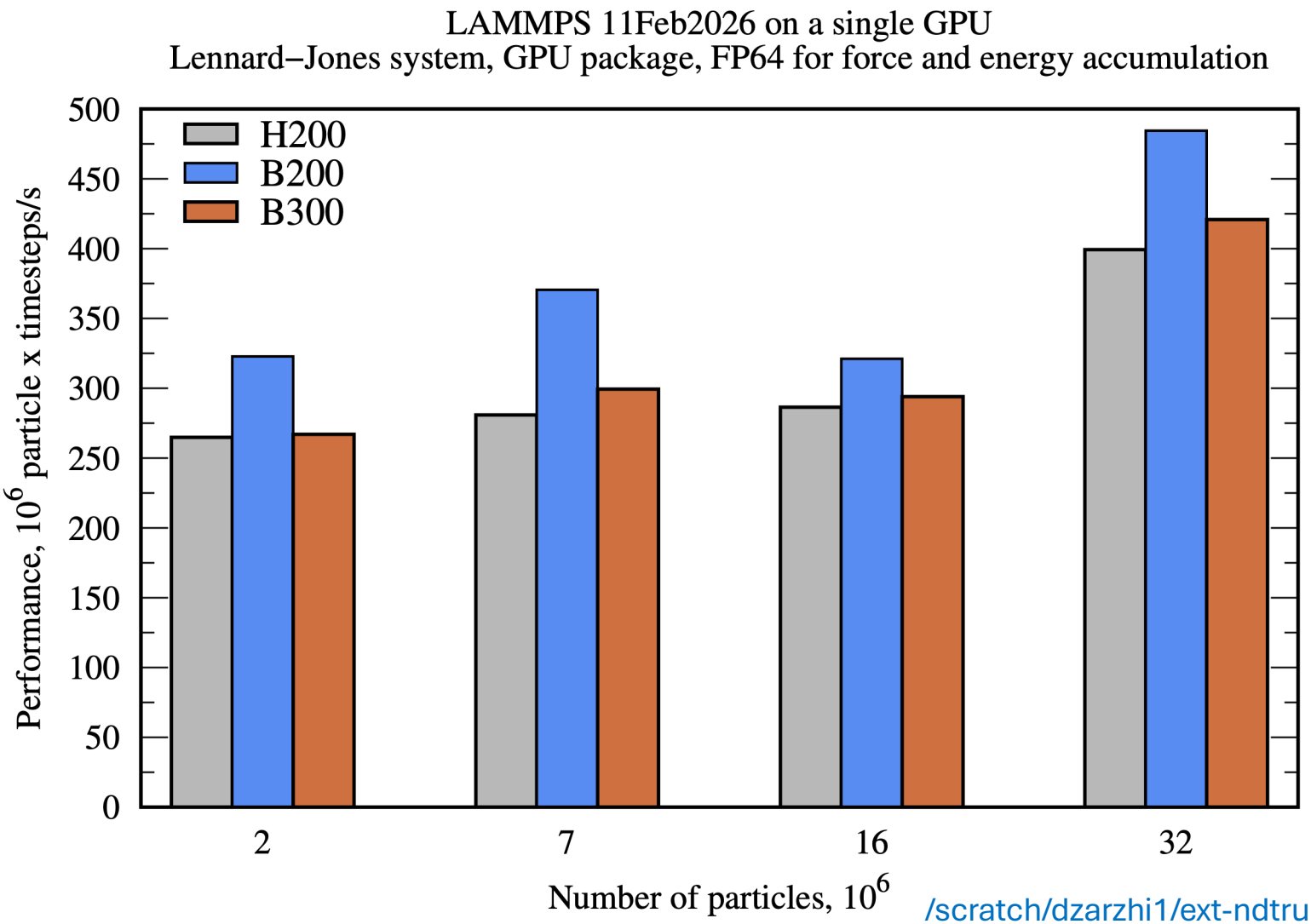
Test	Actual	Min Expected	Max Expected	Status
All-Reduce 10 MB on 4 GPUs	270.05	150.00	190.00	✓ ABOVE
All-Reduce 100 MB on 4 GPUs	519.60	150.00	190.00	✓ ABOVE
All-Reduce 500 MB on 4 GPUs	577.88	150.00	190.00	✓ ABOVE
P2P Transfer 10 MB (GPU 0 <-> GPU 1)	206.74	1200.00	1600.00	⚠️ BELOW
P2P Transfer 100 MB (GPU 0 <-> GPU 1)	451.29	1200.00	1600.00	⚠️ BELOW
P2P Transfer 500 MB (GPU 0 <-> GPU 1)	617.69	1200.00	1600.00	⚠️ BELOW



B300 performs relatively well

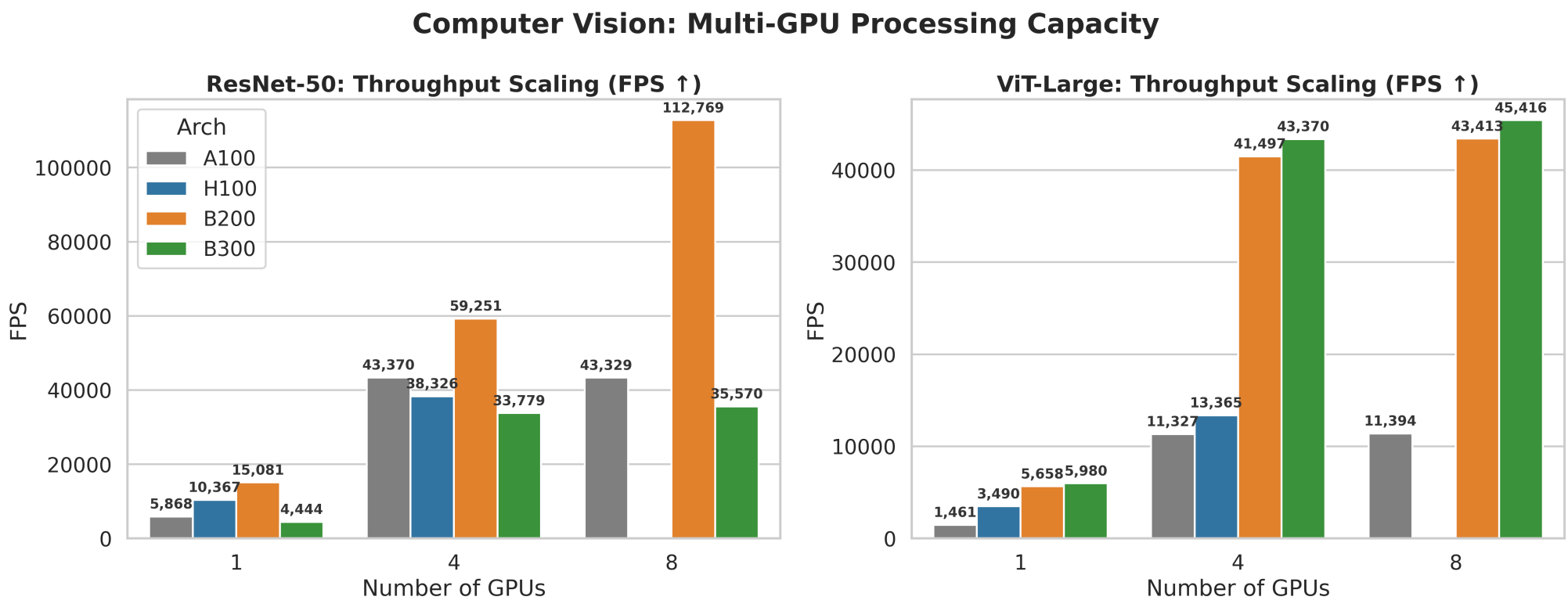
RCC Beagle3 and Midway3

/scratch/dzarzhi1/ext-ndtrung/gromacs/bench/STMV

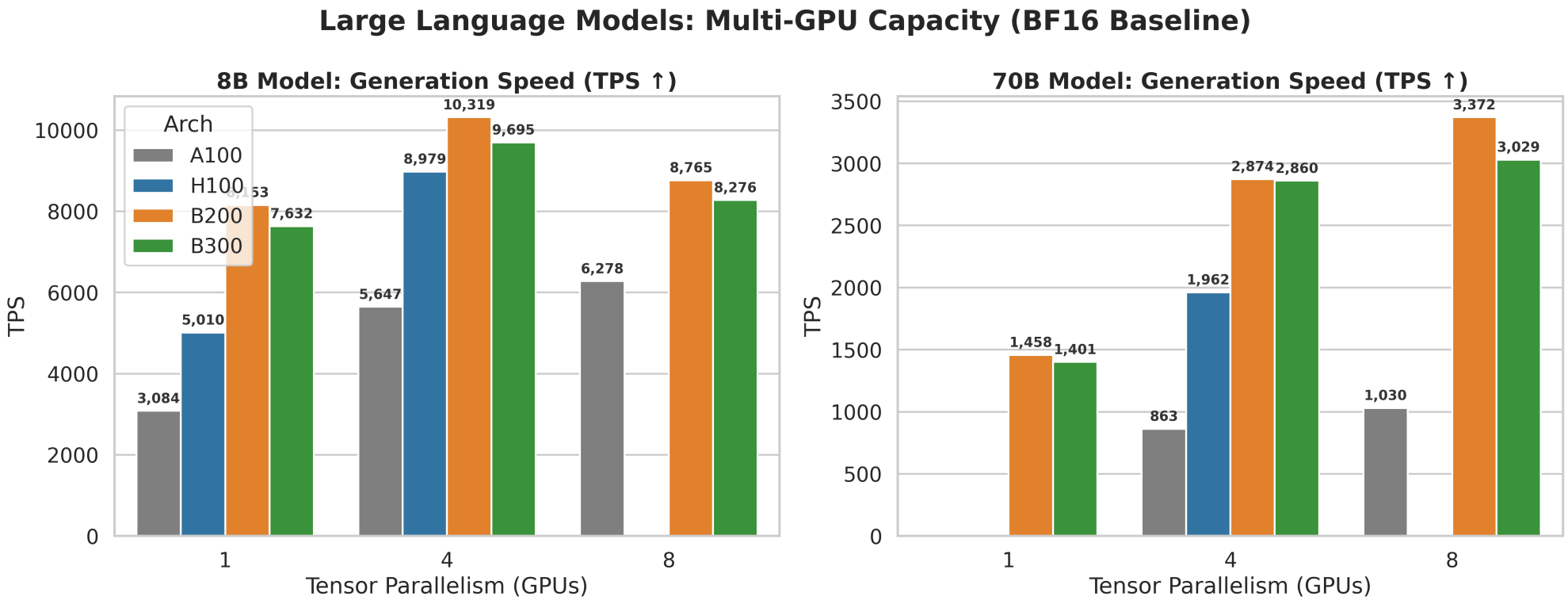


B300 performs surprisingly well with FP64

/scratch/dzarzhi1/ext-ndtrung/lammps/bench

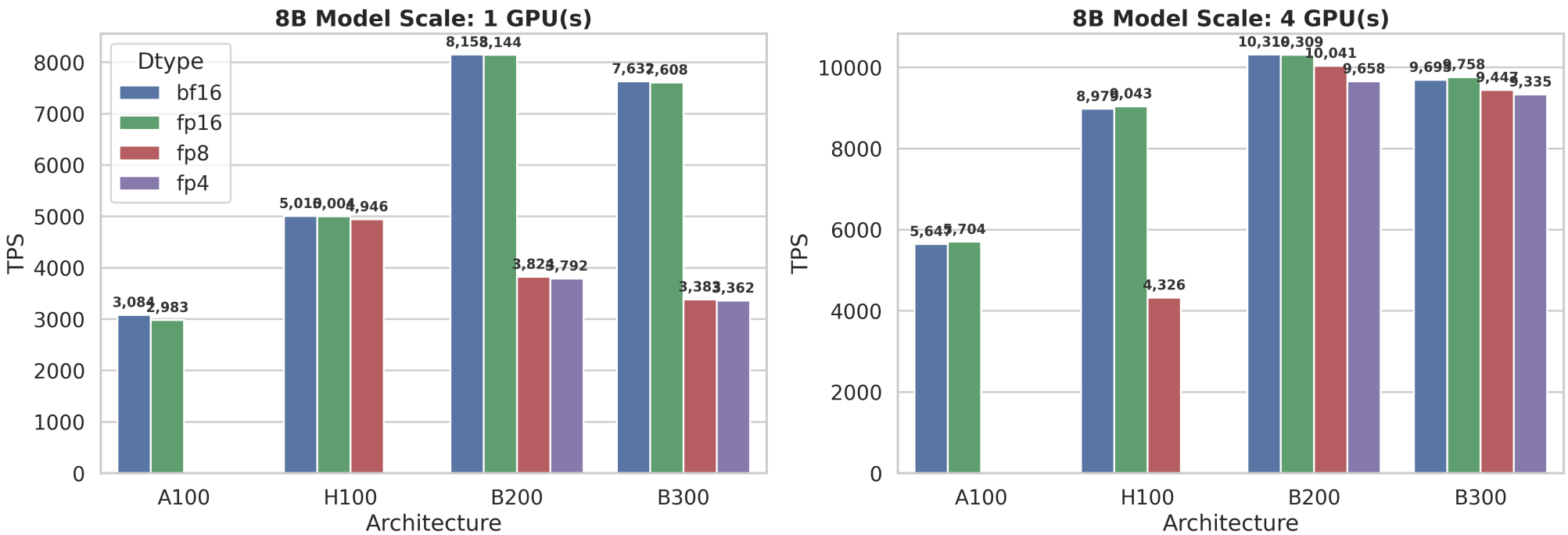


- The chart visually shows that Blackwell (B200/B300) delivers higher performance vs. older GPUs for memory-intensive Vision Transformers (ViT).
- While B200 achieves clear gains on legacy convolutional models (ResNet), B300 struggles probably due to early-stage software or configuration bottlenecks.



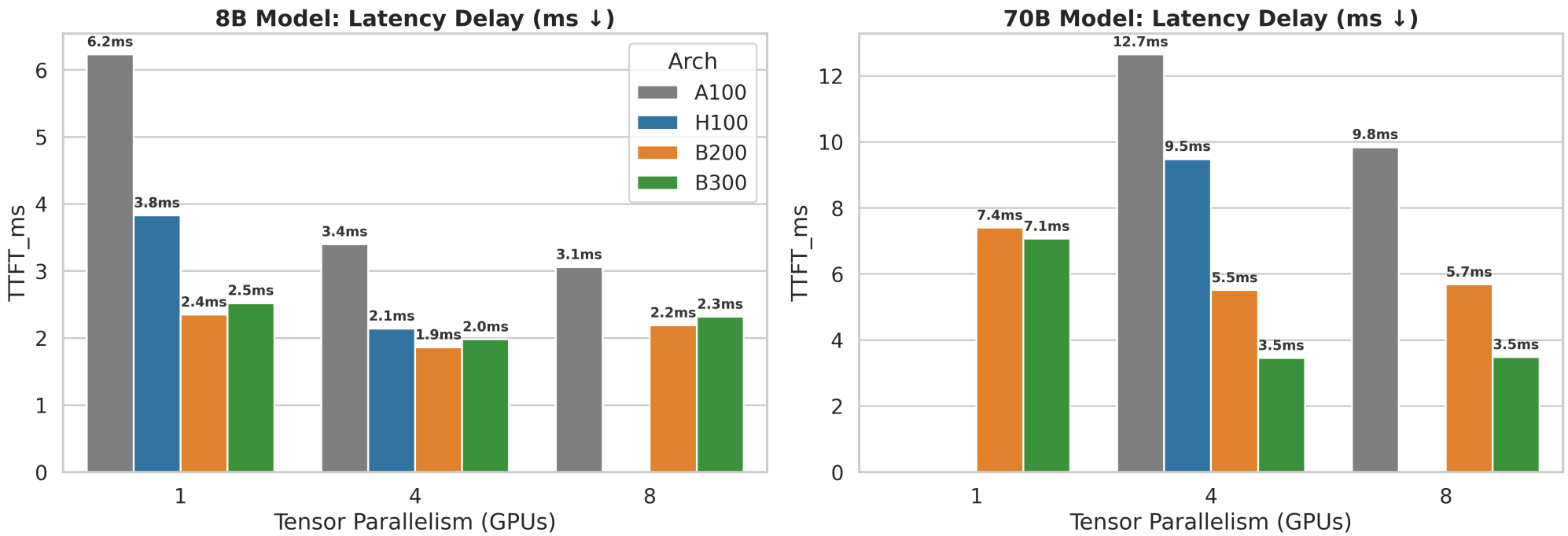
- This chart shows two distinct architectural results: First, Blackwell's higher memory density can allow for model consolidation (the B300's memory density allows us to pack 70B models onto single cards).
- Second, it appears to demonstrate "Over-Sharding Penalty"; showing that while adding GPUs increases raw compute, we wonder if the network communication overhead eventually choke smaller models.

Unlocking the Tensor Engine: The FP8/FP4 Multiplier (TPS ↑)



- While the baseline precision (BF16) proves the new Blackwell silicon is potentially fast, the lower-precision runs (FP8/FP4) expose that the employed open-source inference software (like vLLM) has not yet been optimized for the Blackwell's new hardware-level Transformer Engines.
- There is a drop in FP8/FP4 performance for the 1 GPU runs.

Responsiveness: Time to First Token (TTFT, Lower is Better)



- With older architectures, we had to throw entire 4-GPU nodes at a problem just to get response times down to an acceptable level. The B300 node allows us to achieve instantaneous, sub-4-millisecond responsiveness natively. This means we can serve real-time AI applications using a fraction of the hardware footprint.